Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

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Part I

Problem Framing

1 The Physical Problem

Each patient's micro-electrocorticography (μECoG) array is a **sparse spatial sample** of neural activity over the cortical surface during speech production. The array is placed subdurally over left ventral sensorimotor cortex (vSMC) during surgery, and its position in standardized brain space (MNI-152) is determined post-hoc via electrode reconstruction.

1.1 The cortical surface, not free 3D space

The cortex is a folded two-dimensional sheet. Let $\mathcal{S}$ denote the cortical surface embedded in $\mathbb{R}^3$. The high-gamma analytic amplitude (HGA, 70–150 Hz) is a function on this surface:

$$a : \mathcal{S} \times \mathbb{R} \to \mathbb{R} \tag{1}$$

For patient i with N_i electrodes at cortical surface positions $\{\mathbf{p}_j^{(i)}\}_{j=1}^{N_i} \subset \mathcal{S}$, we observe:

$$d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t) \tag{2}$$

In practice, we approximate surface positions with 3D MNI coordinates. Euclidean distance in MNI does not equal geodesic distance on the cortical surface—electrodes on opposite sulcal banks may be close in 3D but functionally distant. For μECoG arrays on exposed cortex, the local patch is approximately planar, so within-array Euclidean distances are reasonable.

The decoding problem. Given $\{d_j^{(i)}(t)\}$ at approximately known positions, infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$.

The cross-patient problem. Train on patients $\{1, \ldots, K\}$, decode for held-out patient $K{+}1$ whose electrodes sample the surface at different positions.

1.2 What makes this hard

Factor	Quantity	Implication
Electrodes per patient	63–201	Sparse sampling
Electrode spacing	∼2 mm	Finer than registration accuracy
Array offset across patients	15–25 mm	Different cortical regions sampled
Correspondence uncertainty	Several mm	Coordinate alignment is approximate
Tumor mass effect (6/12 patients)	Variable	Macroscopic cortical warping
Trials per patient	46–178	Small-data regime
Total utterance per patient	∼1 min	Insufficient for per-patient SSL

2 The Shared Dynamics Hypothesis

We decompose:

$$\boxed{a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t)} \tag{3}$$

- $f(\cdot, t; \mathbf{y})$: **shared neural field**—the spatiotemporal pattern that $\mathbf{y}$ produces in canonical cortical space.
- $T_i : \mathcal{S}_i \to \mathcal{S}_{\text{canonical}}$: **patient-specific spatial transform**. MNI normalization estimates T_i, imperfectly.
- $r^{(i)}$: **patient-specific residual**—fine-grained neural organization, signal characteristics, pathology.

2.1 Evidence hierarchy

2.1.1 Externally established

Per-patient input + shared backbone is optimal (Singh 2025, N=25, PER 0.49 vs. 0.57; Willett 2023, per-day affine + shared GRU). Articulatory somatotopy exists in vSMC (Metzger 2023, Bouchard 2013). MNI coordinates enable cross-patient ECoG transfer (Chen 2025 SwinTW, 43+9 subjects, PCC 0.765). Motor cortex manifold dynamics preserved across individuals (Safaie 2023) and stable over time (Gallego 2020). Articulatory features robustly encoded in vSMC (Wu 2025).

2.1.2 Internally observed

LOPO achieves PER 0.762 on S14 (vs. per-patient 0.737). Articulatory bottleneck head improves cross-patient decoding. Multi-scale temporal encoding (stride 3+5+10) is the best architectural improvement (0.766 → 0.750). 55 experiments converge to PER 0.750–0.780. All domain adaptation methods (DANN, CORAL, CCA) provide zero benefit.

2.1.3 Working hypotheses

The ∼0.76 convergence partly reflects spatial alignment limitations (not isolated from data scarcity or S14-specific effects). MNI coordinates will improve decoding (depends on registration accuracy and whether alignment is binding). Cross-task pooling (Duraivel 2025) is a high-value, low-risk path to more data.

3 The Neural Manifold Connection

3.1 Two views, one bridge

The cortical field model and the neural manifold hypothesis are dual descriptions. The bridge is the **latent articulatory state** $\mathbf{z}(t) \in \mathbb{R}^k$:

$$\boxed{d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + \epsilon_j^{(i)}(t)} \tag{4}$$

where $g : \mathcal{S} \times \mathbb{R}^k \to \mathbb{R}$ is a **shared observation function**—the somatotopic map. Electrodes over lip motor cortex respond to lip-state variables; electrodes over tongue cortex to tongue-state variables.

The dynamics on the manifold are phoneme-determined:

$$\dot{\mathbf{z}}(t) = h\big(\mathbf{z}(t);\ \mathbf{y}\big) \tag{5}$$

3.2 The factorization is the key insight

Combining the shared dynamics hypothesis with the manifold hypothesis: the shared field $f(\mathbf{p}, t; \mathbf{y})$ factorizes as $g(\mathbf{p}, \mathbf{z}(t))$, separating space from time through a low-dimensional bottleneck. Instead of learning an arbitrary spatiotemporal function, the model learns:

1. A spatial function g: how each brain position encodes articulatory features (the somatotopic map).
2. A dynamical system h: how the articulatory state evolves during speech.

3.3 Articulatory manifold dimensionality

The manifold has physical meaning. For our 9 phonemes:

Contrast	**Phonemes distinguished**	**Dim.**
Place of articulation	labial (b,p,v) vs. velar (g,k)	$\sim$2
Manner	stop vs. fricative vs. vowel	$\sim$2
Voicing	voiced (b,g,v) vs. voiceless (k,p)	$\sim$1
Vowel height/backness	(a,æ,i,u) contrasts	$\sim$2
Total effective dimension		**$\sim$5–8**

This is far below electrode count (63–201): the decoding problem is well-posed.

3.4 Cross-patient transfer as manifold alignment

Each patient's array gives a different projection of the shared manifold: $\mathbf{d}^{(i)}(t) \approx W^{(i)}\mathbf{z}(t) + \boldsymbol{\epsilon}^{(i)}$, where $W^{(i)} \in \mathbb{R}^{N_i \times k}$ is determined by electrode positions. MNI coordinates provide the geometric relationship between these projections a priori, rather than requiring post-hoc alignment.

4 Decomposing Cross-Patient Variation

4.1 The spatial transform

$$T_i(\mathbf{p}) = A_i \cdot \mathbf{p} + \mathbf{b}_i + \delta_i(\mathbf{p}) \tag{6}$$

A_i: rigid/affine (corrected by MNI normalization). $\mathbf{b}_i$: translation (15–25 mm array offsets). δ_i: nonlinear residual including **tumor mass effect**—a geometric mapping error that physically displaces the somatotopic map.

4.2 Inter-subject correspondence uncertainty

Not a fixed constant. Sources: native-space localization (sub-mm), brain shift correction (variable), atlas warping (dominant). Net result: several mm for healthy anatomy, more near tumors.

4.3 Resolution mismatch ⇒ dual spatial encoding

Scale	**Resolution**	**Best encoded by**	**What it captures**
Cross-patient	$\sim$10–15 mm	MNI coordinates	Articulatory somatotopy
Intra-patient local	$\sim$2 mm	Grid topology	Fine-grained spatial patterns

5 What Our Experiments Reveal

Evaluation recipe dominates: training improvements ∼7 pp, evaluation improvements ∼6.5 pp (weighted k-NN, TTA, articulatory head, best-of per fold). **Multi-scale temporal validated:** stride 3+5+10 is best architecture (−1.6 pp). **SSL carries partial structure:** JEPA + LP-FT achieves 0.797 (vs. 0.854 basic). **The** ∼0.76 **wall:** 55 experiments converge; cannot attribute to a single bottleneck without population-level evaluation.

Part II

The Multi-View Reconstruction Analogy

6 From 3D Pose Estimation to Neural Field Decoding

The cross-patient decoding problem has a precise structural analog in computer vision: **fitting a parametric 3D model to 2D observations from multiple viewpoints**. This analogy, inspired by holistic performance capture approaches (Hewitt et al., 2024, arXiv:2410.11520), motivates the reprojection loss that is central to our architecture.

6.1 The pose estimation setup

In 3D human pose estimation (e.g., fitting SMPL to 2D keypoints):

1. A **parametric 3D model** (SMPL) encodes human body structure, parameterized by pose θ and shape β.
2. A **camera** projects 3D model vertices to 2D image coordinates via known projection function π.
3. We observe **2D keypoint detections** $\{x_j\}$—sparse, noisy projections of the 3D model.
4. The **reprojection loss** $\sum_j \|x_j - \pi(V_j(\theta, \beta))\|^2$ drives optimization of pose and camera parameters.

6.2 The mapping

3D Pose Estimation		**Neural Field Decoding**
3D body model (SMPL mesh)	↔	Shared neural field $g(\mathbf{p}, \mathbf{z})$ on cortex
Pose parameters θ (joint angles)	↔	Articulatory state $\mathbf{z}(t)$ (vocal tract config.)
Shape parameters β (body proportions)	↔	Patient embedding $e^{(i)}$ (neural residual)
Camera extrinsics $[R \mid t]$	↔	MNI offset $\Delta^{(i)}$ (registration correction)
Camera intrinsics K	↔	Per-patient gain/bias (signal scaling)
2D keypoint observations x_j	↔	Electrode observations $d_j^{(i)}(t)$
Projection function $\pi(\cdot)$	↔	Observation function $g(\cdot)$
Keypoint visibility/occlusion	↔	Dead/masked electrodes
Reprojection loss $\|\pi(V) - x\|^2$	↔	**Reprojection loss** $\|g(\mathbf{p}+\Delta, \hat{\mathbf{z}}) - d\|^2$

6.3 The multi-view extension

The analogy deepens when we consider multi-patient training as **multi-view 3D reconstruction**:

Multi-View Reconstruction		**Multi-Patient Neural Field**
Multiple cameras observe the same 3D scene from different viewpoints	$\leftrightarrow$	Multiple patients "observe" the same cortical dynamics from different electrode positions
Each camera has own extrinsics $[R_i \mid t_i]$	$\leftrightarrow$	Each patient has own offset $\Delta^{(i)}$, embedding $e^{(i)}$
3D structure is **shared**; only viewpoints differ	$\leftrightarrow$	Neural field g is **shared**; only electrode positions differ
Joint optimization: 3D structure + all camera params	$\leftrightarrow$	Joint optimization: neural field + all patient params
More cameras = better reconstruction	$\leftrightarrow$	More patients = better field estimate
Novel view synthesis: render from new camera	$\leftrightarrow$	Novel patient: predict from new electrode positions

6.4 The NeRF connection

Our observation decoder is a **cortical neural field**—structurally analogous to a Neural Radiance Field (NeRF):

NeRF		**Neural Field Perceiver**
Input: (x, y, z, θ, ϕ)	$\leftrightarrow$	Input: $(\mathbf{p}_{\text{MNI}}, \mathbf{z}_{\text{articulatory}})$
Output: (color, density)	$\leftrightarrow$	Output: HGA activity (scalar)
MLP: position $\rightarrow$ features	$\leftrightarrow$	$\text{MLP}_{\text{spatial}}$: position $\rightarrow \phi$ (spatial tuning)
Train: render from known cameras, compare to images	$\leftrightarrow$	Train: "render" at known electrodes, compare to activity
Novel view: render from new pose	$\leftrightarrow$	Novel patient: predict from new positions

The key difference: NeRF uses an unconstrained MLP. Our bilinear factorization $\hat{d} = \phi(\mathbf{p})^\top \psi(\mathbf{z})$ is a **structured** neural field that separates spatial tuning from temporal activation—physically motivated by the separability of somatotopic organization and articulatory dynamics.

6.5 What the analogy reveals

Why previous cross-patient methods failed:

- **CCA/Procrustes/CORAL** are like aligning camera views by matching 2D feature statistics. This works for similar viewpoints but fails when viewpoints are very different (15–25 mm array offsets = cameras at very different positions).
- **Per-patient SpatialConv** is like having a separate 3D model per camera—no information sharing between views.
- **Our approach** is like **Structure from Motion**: jointly estimate shared 3D structure and all camera parameters from multi-view observations. The reprojection loss is the standard SfM objective. MNI coordinates are like having approximate GPS for each camera—not perfect, but enough to initialize optimization.

Why reprojection loss is better than arbitrary SSL:

- Previous SSL (JEPA, VICReg, masked reconstruction) used *arbitrary* reconstruction targets with no physical meaning.
- The reprojection loss is *physically grounded*: "if your latent state estimate is correct, you should be able to predict what every electrode observes at its known brain position."
- This is the neural analog of: "if your 3D pose estimate is correct, the projected keypoints should match the observed 2D keypoints."
- The loss is **self-supervised**—no phoneme labels needed—but **geometrically constrained** by the observation model.

Part III

Architecture Design

7 Design Principles

Every component maps to the observation function (4):

Subproblem	Mathematical operation	Module
Invert g: observations $\to$ latent state	Set-to-fixed projection + geometric prior	Spatial Encoder
Implement g: latent + position $\to$ prediction	Bilinear neural field	Observation Decoder
Model dynamics h: temporal trajectory	Multi-scale sequence modeling	Temporal Encoder
Classify trajectory $\to$ phonemes	Per-position classification	Phoneme Decoder

8 Spatial Encoder (∼100K params)

8.1 Electrode tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \to d)}_{\text{activity}} + \underbrace{\text{Linear}(\mathbf{p}_j + \Delta^{(i)} \to d)}_{\text{MNI PE}} + \underbrace{\text{Linear}(\text{row}_j, \text{col}_j \to d)}_{\text{Grid PE}} + \underbrace{e^{(i)}}_{\text{patient}} \tag{7}$$

Per-patient parameters ($3 + d = 67$ per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: learnable MNI offset (registration correction), initialized to $\mathbf{0}$.
- $e^{(i)} \in \mathbb{R}^d$: learnable patient embedding, initialized $\sim \mathcal{N}(0, 0.02)$.

Dead electrodes are excluded (not masked). Variable N_i is handled natively by attention.

8.2 Perceiver cross-attention with distance bias

$L = 16$ **virtual electrodes** with learnable MNI positions $\{(\text{v}_x, \text{v}_y, \text{v}_z)_i\}$:

$$\text{query}_i = \text{embed}_i + \text{MNI_PE}(\text{v_pos}_i) \tag{8}$$

Cross-attention with spatial locality prior:

$$A_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d}} \; - \; \alpha \cdot \|\text{v_pos}_i - (\mathbf{p}_j + \Delta^{(i)})\|^2 \tag{9}$$

where $\alpha > 0$ is learnable (initialized 0.1), controlling spatial blur: large α = sharp attention (precise MNI), small α = fuzzy (robust to misalignment).

4-head attention, optional 1-layer self-attention, then **MaxPool** across L virtual outputs $\rightarrow s(t) \in \mathbb{R}^d$.

9 Observation Decoder (∼25K params)

The forward observation model—the "renderer" in the pose estimation analogy:

$$\boxed{\hat{d}_j(t) = \phi(\mathbf{p}_j + \Delta^{(i)},\; e^{(i)})^\top \; \psi(s(t))} \tag{10}$$

- $\phi = \text{MLP}_{\text{spatial}}(\mathbf{p}, e) \in \mathbb{R}^k$: spatial tuning vector at position $\mathbf{p}$ for patient i.
- $\psi = \text{MLP}_{\text{latent}}(s) \in \mathbb{R}^k$: articulatory activation for estimated state $s(t)$.
- $k = 32$: bilinear interaction dimensionality.

Reprojection loss (self-supervised, no labels):

$$\mathcal{L}_{\text{reproj}} = \frac{1}{N_i} \sum_{j=1}^{N_i} \left\| d_j(t) - \hat{d}_j(t) \right\|^2 \tag{11}$$

This loss trains the encoder, decoder, $\Delta^{(i)}$, and $e^{(i)}$ simultaneously—exactly as reprojection error trains pose, camera params, and shape in 3D fitting.

Electrode masking: During training, drop 20–40% of electrodes from encoder input; keep them in reprojection loss. The model must spatially interpolate through the neural field—geometric masked reconstruction.

10 Temporal Encoder (∼70K params)

10.1 Multi-scale tokenization (experimentally validated)

$$\begin{aligned} s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=3,s=3)} T_3 \approx 67 \text{ tokens at } \sim 67\,\text{Hz} \\ s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=5,s=5)} T_5 \approx 40 \text{ tokens at } \sim 40\,\text{Hz} \\ s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=10,s=10)} T_{10} \approx 20 \text{ tokens at } \sim 20\,\text{Hz} \end{aligned} \tag{12}$$

Concatenate all ≈127 tokens with scale PE + sinusoidal temporal PE.

10.2 Temporal transformer

2 layers, 4 heads, d=64, FFN dim 256. Each token attends across all scales and times. The 67 Hz token at t=100 ms can attend to the 20 Hz token at t=800 ms.

11 Phoneme Decoder (∼5K params)

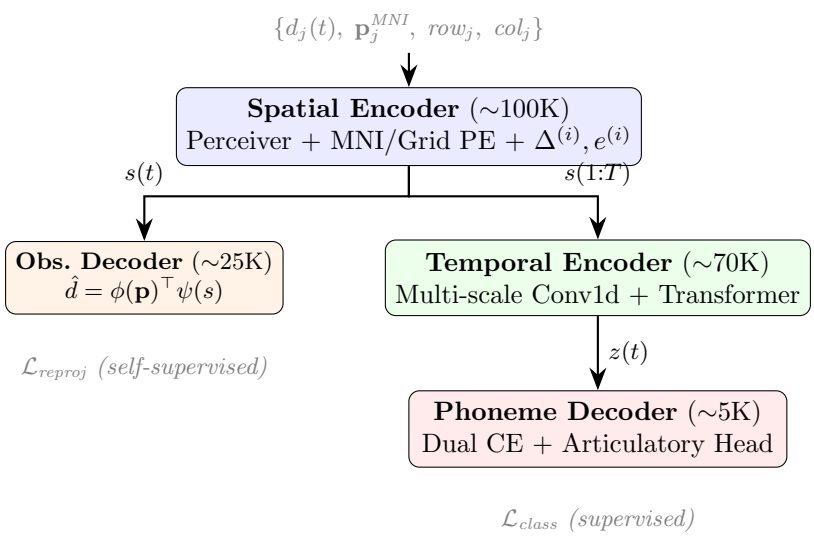
Dual head (experimentally validated):

$$\text{logits}_p = \frac{1}{2}\Big(\underbrace{\text{Linear}(z \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \text{normalize}(\text{Linear}(z \to 15)) \cdot A^\top}_{\text{articulatory head}}\Big) \tag{13}$$

where $A \in \mathbb{R}^{9\times 15}$ is the fixed articulatory feature matrix and τ is a learnable temperature.

Classification loss: $\mathcal{L}_{\text{class}} = \frac{1}{3}\sum_{p=1}^{3} \text{FocalCE}(\text{logits}_p, y_p;\ \gamma{=}2, \text{LS}{=}0.1)$ with mixup (α=0.2).

12 Complete System



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \cdot \mathcal{L}_{\text{reproj}}} \tag{14}$$

Module	**Params**	**Role**
Spatial encoder	∼100K	Invert observation function
Observation decoder	∼25K	Forward model (reprojection)
Temporal encoder	∼70K	Dynamics modeling
Phoneme decoder	∼5K	Classification
Total shared	**∼200K**	
Per-patient (×10)	670	Registration + identity
Grand total	**∼201K**	(cf. current pipeline: 122K)

Part IV

Training, Evaluation, and Next Steps

13 Training Pipeline

13.1 Phase 1: Source patient training (supervised + reprojection)

9 source patients, ∼1315 labeled trials. Each batch:

1. Spatial encoding: construct tokens, mask 20–40% electrodes, Perceiver $\rightarrow s(t)$.
2. Reprojection: decoder predicts all electrodes (including masked); $\mathcal{L}_{\text{reproj}}$.
3. Temporal encoding: $s(1{:}T) \rightarrow$ multi-scale $\rightarrow$ transformer $\rightarrow z(t)$.
4. Classification: dual head $\rightarrow \mathcal{L}_{\text{class}}$.
5. Total: $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \mathcal{L}_{\text{reproj}}$ (λ: 0.5 $\rightarrow$ 0.1 annealed).

AdamW, differential LR (shared 1e-3; virtual positions/patient Δ 3e-4), cosine schedule, warmup 20 epochs, early stopping patience 7.

Optional: alternate labeled batches (both losses) with unlabeled batches from continuous recordings or Lexical patients (reprojection only).

13.2 Phase 2: Target adaptation (LP-FT)

Per CV fold on S14 (153 trials, 5-fold grouped-by-token):

1. **Linear probe**: freeze encoder, train decoder + $\Delta^{(\text{S14})} + e^{(\text{S14})}$.
2. **Fine-tune**: unfreeze encoder at 0.1× LR.

13.3 Phase 3 (optional): Reprojection pre-training

Pre-train spatial encoder + observation decoder on all available data (continuous recordings + Lexical patients) using $\mathcal{L}_{\text{reproj}}$ only. No labels needed. Then initialize Phase 1 from pre-trained weights.

14 Evaluation

PER via edit distance, grouped-by-token 5-fold CV. Full recipe: weighted k-NN (k=10, cosine), TTA 16, dual head, best-of per fold, source k-NN at 0.5× weight.

14.1 Key ablations

15 Data Paths

Highest priority (lowest risk): Cross-task pooling from Duraivel 2025—same tokens, same phonemes, overlapping patients, compatible preprocessing.

Medium priority: MNI coordinate sourcing (BioImage Suite / Brainlab / ECoG_Recon on Box). Hard dependency for the Perceiver architecture.

ID	Ablation	Question
A1	MNI PE present/absent	Do MNI coordinates help? (central question)
A2	All patients vs. S14 only	Is the 0.76 wall S14-specific?
A3	Δ / e / both / neither	Which per-patient params matter?
A4	$L \in \{4, 8, 16, 32\}$	How many virtual electrodes?
A5	$\lambda \in \{0, 0.01, 0.1, 0.5\}$	Does reprojection loss help?
A6	+ Duraivel cross-task data	Does more data help? (independent of architecture)

Lower priority: Continuous recording extraction + VAD curation for Phase 3 SSL.

16 Success Criteria

Level	Criterion
Minimum	A1: Perceiver + MNI outperforms Perceiver without MNI by $\geq$2 pp on S14
Target	PER < 0.740 on S14 (exceeds per-patient supervised 0.737)
Strong	Population-level improvement across all patients
Interpretability	Virtual electrodes cluster over vSMC; $\Delta^{(i)}$ larger for tumor patients; $\phi(\mathbf{p})$ recovers somatotopy

17 Summary

This document formalizes cross-patient speech decoding as **multi-view neural field reconstruction**: multiple patients observe the same cortical dynamics from different electrode positions, analogous to multiple cameras observing the same 3D scene. The shared observation function $g(\mathbf{p}, \mathbf{z})$ is the somatotopic map; patient-specific parameters (Δ, e) are camera calibration; the reprojection loss trains spatial alignment without labels.

The Neural Field Perceiver implements this framework: Perceiver cross-attention with MNI distance bias inverts the observation function, a bilinear decoder implements it forward for reprojection, multi-scale temporal transformer models articulatory dynamics, and per-position classification decodes the phoneme sequence. The architecture uses $\sim$200K shared parameters and 67 per patient, directly mapping every component to the observation equation.